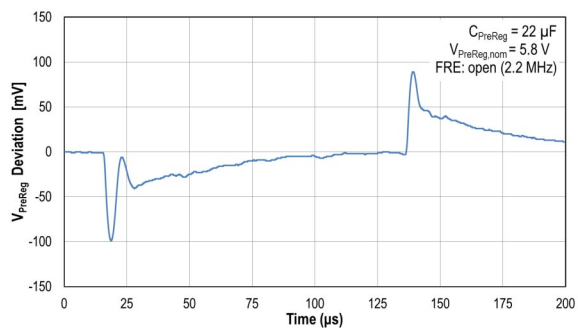
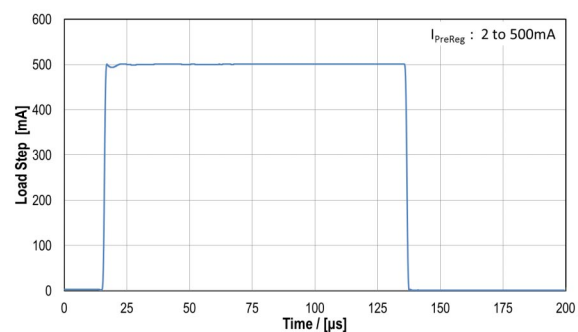
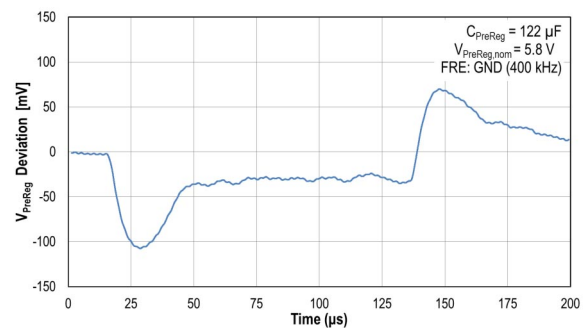
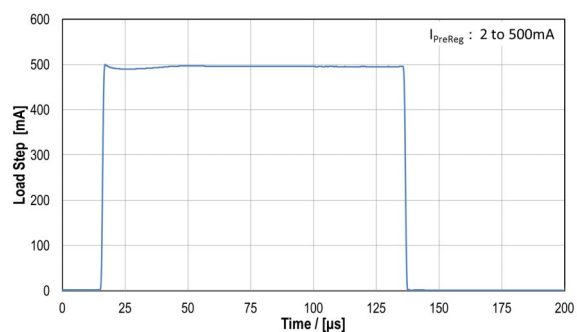


PreReg Dynamic Load Response (2mA to 500mA)
(FRE-Pin: open ; $V_{\text{PreReg,nom}} = 5.8 \text{ V}$)



PreReg Dynamic Load Response (2mA to 500mA)
(FRE-Pin: GND ; $V_{\text{PreReg,nom}} = 5.8 \text{ V}$)



6.4 Frequency setting

6.4.1 Introduction

The frequency source supplies the step up pre regulator and the step down pre regulator with a constant frequency. The synchronous power switches of the step down pre regulator will switch directly with the frequency f_{osc} .

The frequency range of the step down pre regulator can be set to the high switching frequency range by leaving pin FRE open or to the low switching frequency range by connected the pin FRE to GND. The switching frequency will be set to the default value of the chosen frequency range. Optionally it can be fine tuned by SPI command ([BCK_FREQ_CHANGE](#)) or the spread-spectrum option can be activated ([BCK_FRE_SPREAD](#)).

The switching frequency range of the step up pre regulator is lower than the switching frequency range of the step down pre regulator.

The switching frequency of the step down pre regulator is offered at pin SYN for the optional external switch mode post regulator for the μC core supply, if the SEC pin is left open.

The synchronization function is not available in PFM mode.

The TLF35584 cannot be synchronized to an external frequency source.

6.4.2 Electrical characteristics frequency setting

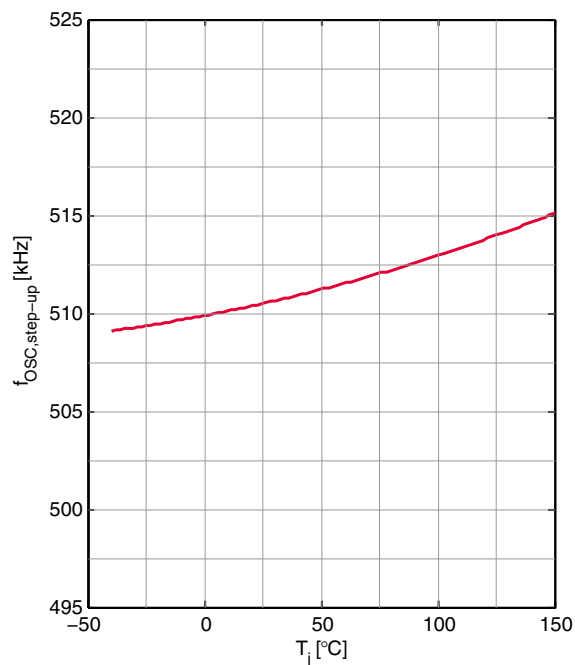
Table 9 Electrical Characteristics: Oscillator

$V_{VS} = 6.0 \text{ V to } 40 \text{ V}$, $T_j = -40^\circ\text{C to } +150^\circ\text{C}$, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

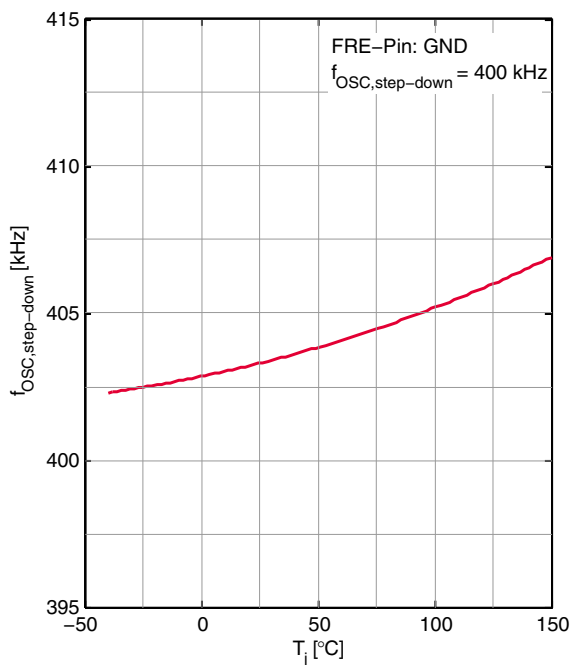
Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Frequency Setting FREQ							
Step-up frequency range	$f_{\text{OSC,step-up}}$	400	500	600	kHz	–	P_6.4.2.1
Step-down low frequency range	$f_{\text{OSC,step-down}}$	300	400	500	kHz	FRE pin connected to GND	P_6.4.2.2
Step-down high frequency range	$f_{\text{OSC,step-down}}$	2000	2200	2500	kHz	FRE pin open	P_6.4.2.3

6.4.3 Typical Performance Characteristics

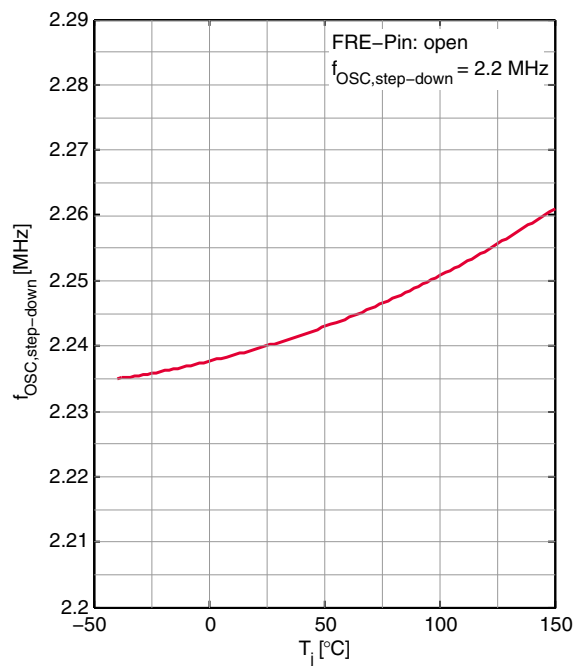
Step-up Switching Frequency $f_{\text{OSC,step-up}}$ versus Junction Temperature T_j (FRE-Pin: open)



Step-down Switching Frequency $f_{\text{OSC,step-down}}$ versus Junction Temperature T_j (FRE-Pin: GND)



Step-down Switching Frequency $f_{\text{OSC,step-down}}$ versus Junction Temperature T_j (FRE-Pin: open)



7 Post Regulators

7.1 Introduction

The TLF35584 includes a number of linear low drop post regulators and trackers and offers the possibility to connect an external post regulator for the μ C core supply if needed.

The linear post regulators and trackers are supplied from pins FBx. The band gap 1 for regulator block provides the reference values for the error amplifiers of μ C supply LDO (pin QUC), the communications supply LDO (pin QCO) and the reference voltage source (pin QVR). The trackers get their reference value from the reference voltage source (pin QVR). The output voltage of trackers 1 and 2 (present at pins QT1 and QT2) is following the reference voltage source Volt_Ref with a small drop.

An additional external post regulator can be added to deliver the core supply for the micro processor. If this option is used the configuration pin SEC must be left open. If the option is not used, pin SEC must be connected to ground.

The post regulator is to be connected external and uses its own reference voltage, the input is fed from the pre regulator output voltage V_{PREREG} (which is similar to the values at pins FBx). The post regulator is enabled by a high signal at pin EVC and switched off with a low signal at pin EVC. A synchronization signal (in phase or shifted by 180 degree with the step down pre regulator signal) is offered at pin SYN for usage of a switch mode post regulator.

All output voltages of the post regulators are connected to the voltage monitoring function (please refer to chapter [Monitoring Function](#)).

In case of an over voltage the related post regulator will be switched off, the shutdown signal is generated by the voltage monitoring function.